

The mathematics behind my Turn-Based game project

What is this project about

I wanted to make a project where I am able to integrate math together with my code - something which I have not done. Hence, I decided to make a turn-based game project that has three different options:

1. **Attack** – Deals damage to the opponent to lower their health
2. **Heal** – Restores a portion of the player's own health
3. **Train** – Permanently levels up the player's Attack or Heal skills for the remainder of the game, hence increasing the effectiveness of each option

The defining feature of this project is that it relies on probability sequences rather than uniform randomness, hence making the game more interesting. For example, a level 0 attack does not randomly pick any number from 0 to 5. It instead employs a weighted distribution where certain damages (i.e. middle numbers) are more favoured than other extreme outcomes such as 0 or 5.

This writeup will cover the mathematics behind the game. Additionally, this will also cover the logic behind the code and how I used the module “Textual¹” to further enhance the user experience of the game.

¹ “Textual - Home.” *Textual Documentation*, textual.textualize.io/.

The mathematics behind the game

This section details the mathematical framework behind the game's mechanics. There are several issues that arise when trying to make this game:

1. Player 1 will always have an advantage over player 2 if both players are equal in skill level ($s_1 = s_2$).
2. A heal option that is equivalent to the damage function in expected value would result in a statement. This is shown in the payoff matrix (an example of a scenario where the average health healed is +2 and damage dealt is -2) which highlights that the dominant strategy for player 1 is to attack followed by player 2 healing themselves thereby locking the game into a (Attack, Heal) Nash Equilibrium.

		PLAYER 2	
		Attack	Heal
PLAYER 1	Attack	(-2, -2)	(0, 0)
	Heal	(-2, 0)	(0, 0)

3. Having an option to skill up might result in a situation whereby the players would repeatedly spam this option in order to get superior attack/heal stats or not use this option at all.

Hence, after considering the following issues, the main aim of this section could be summarised by asking this question: How to create the game whereby the damage and heal output (d and h) is dependent on the skill level of the player (s) and also ensures turn-order neutrality.

This section will go by parts in order to lay out systemically each feature of the game (i.e. the max health, the probability function for damage dealt, the probability function for health healed etc) and also address the issues faced.

Part I: Damage dealt vs Health Regained

For this section, the value of the damage dealt will be represent by d and the value of the health regained will be represent by h . It is observed from above that a stalemate would occur if:

$$E[h] \geq E[d]$$

Hence, a potential solution would be to make $E[d] > E[h]$. However, this would result in the players from refusing to use the heal function all together. Let's consider a hypothetical situation whereby $E[d] = 3$ and $E[h] = 2$. The game payoff below will illustrate the possible moves for the first 2 turns for each player where the first letter is the first move and second letter is the second move (A – Attack, H – Heal) and the final number represents the health of both the players after two turns (Max health is 15):

		Player 1			
Player 2		AA	AH	HA	HH
	AA	(9,9)	(11,12)	(9,12)	(11,15)
	AH	(12,11)	(14,14)	(12,14)	(14,15)
	HA	(12,11)	(12,14)	(12,12)	(12,15)
	HH	(15,13)	(15,14)	(15,14)	(15,15)

In the current model, the max health of the game is 15 and therefore the players will not be able to gain health beyond 15. However, this limits the choices of player 1 in their first turn. Since player 1 starts out in max health, utilising the heal function will have no net positives hence player 1 will always end up choosing the attack function which will reduce player 2's health to 12. Therefore, we can simply ignore columns 3 and 4 of the payoff table and focus on column 1 and 2. Now, player 2 is present with the following two options:

Option 1 (Heal option): *Player 1's health* $\rightarrow 15$ | *Player 2's health* $\rightarrow 14$

Option 2 (Attack option): *Player 1's health* $\rightarrow 12$ | *Player 2's health* $\rightarrow 12$

It is evident that option 2 is more favourable to player 2 as opposed to option 1 and hence player 2 will also choose the attack option. Now, in the second turn of player 1, the player is able to choose between either healing or attacking. Whilst healing would increase their health to 14, it gives player 2 the opportunity to attack leading to player 2 gaining an advantage and player 1 loosing their first-mover advantage. Hence, player 1 is likely to

continue attacking. This results in both players to choose the **AAAA** option thereby not using the heal function at all. In this scenario, player 1 is bound to be more aggressive and will attack on every turn which will be the optimal strategy for player 1. Since player 1 starts first, this strategy would mean that player 2 will reach a health of 0 before player 1 does thereby allowing player 1 to win every game by simply spamming attack.

In the example above I fixed ***d*** to be 3 while ***h*** to be 2 to make it easier to analyse a situation whereby $E[d] > E[h]$. However, the actual game aims to have a range of values for ***d*** and ***h*** with each value having a certain probability of being picked. Therefore, the game must now focus on how to ensure that both heal and damage are being actively used strategically by the players whilst at the same time preventing a stalemate.

Since the game's win condition is only when the health of one of the players reaches 0, attacking must remain as the central focus of the combat loop while healing functions as a supplementary action that is used tactically to gain an edge. To achieve this balance, the following 2 design frameworks for integrating the heal mechanic were employed:

1. Decreasing standard deviation for the range of values of ***h*** as opposed to ***d***
2. Creating a streak multiplier for ***d***
3. Adding a system whereby healing becomes less effective if the other players deals a high ***d*** (Wounding system)

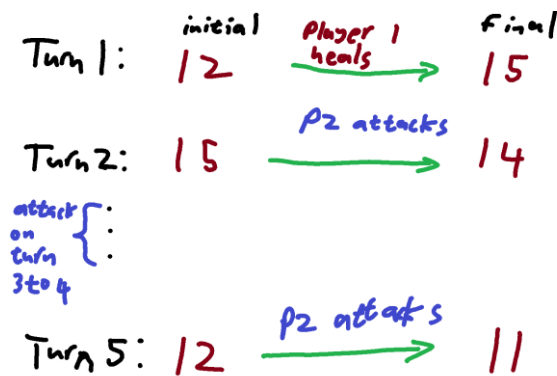
It is possible to make the standard deviation of ***h*** (***Std(h)***) to be lesser than one by clustering all the potential values to be closer to $E[h]$. Therefore, in order to first figure out the ***Std(h)*** that can be used, I need to find the ideal value for $E[h]$. Since $E[h]$ is closely related to $E[d]$, I will also figure out the ideal values for $E[d]$ and $E[h]$ when the player is in level 0. From above it is important for $E[d] > E[h]$ in order for the game to not wind up in a stalemate. However, it is also evident that in such a scenario, the game will simply involve spamming the attack button. Hence a streak multiplier and a wounding system will be used to incentivise the attack option. This ensures the heal function remains strategically useful to the player but also prevents the players from spamming it thus resulting in a stalemate.

1. Deciding the range of values of d :

Since the game has a max health of 15, I decided that the critical damage dealt should be 5 (which is $\frac{1}{3}$ the max health). This will make the game engaging as dealing a critical damage would cause the other player to lose a significant portion of their health. I also decided that 0 should be the minimum damage a player can deal as it will bring some form of uncertainty when deciding to use the attack option. Therefore, for a level 0 player, the range will be $0 \leq d \leq 5$. Since the game will be involving increasing levels, I decided that $E[d]$ should be around 1.

2. Deciding the range of values of h :

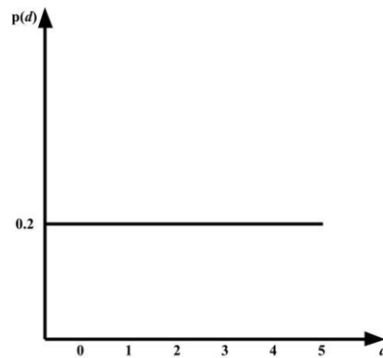
Since I will be using a cooldown mechanism for health, I can let the $E[h]$ be greater than $E[d]$. However, if $E[h] \geq 3$, the game will require at least 4 turns just to reduce the health of the player to be lower than the original health thus drastically increasing the timeline of the game. This is highlighted in the following drawing (Excludes player 1's turn):



And since $E[d] \approx 1$, the most appropriate value for $E[h]$ will be around 2. However, to ensure that the game will be exciting, some levels of uncertainty are required. Therefore, I decided that the appropriate range should be $1 \leq h \leq 3$.

3. Deciding the probabilities for each value of d :

In order to decide the probability of each value of d (0 to 5), I decided to use a probability function. There are various probability functions that can be used. Firstly, I decided to employ discrete random variables (PMF) instead of continuous random variables (PDF). This is to ensure that the damage dealt and the health restored are represented as whole numbers (aka discrete), thereby preventing the complexities of potentially having an infinite range of d when $0 \leq d \leq 5$ (e.g. $d = 0.53932$ or $d = 4.78901$). Hence, d can only have the possibility of being an integer from 0 to 5 and not just any number. This also makes it easier for the players to follow the game without being overloaded with information. The graph below shows an example of discrete random variables for $0 \leq d \leq 5$, where each number has an equal probability of being picked:



From the graph above, it is evident that each value of damage has an equal 20% chance of being picked.

However, for this turn-based game, it is essential that there is some variability in the picking out certain values thus making the game more interesting and unpredictable. The following table highlights the probabilities for each value of d :

d	$P(X = d)$
0	a
1	b
2	c
3	d
4	e
5	f

Hence,

$$a + b + c + d + e + f = 1$$

Additionally, we also decided that $E[d] = 1$. After deciding on using PMF, I then had to figure out what type of distribution to employ. I focused on 4 of the following distribution models: linear distribution, quadratic distribution, reciprocal distribution and exponential distribution. The following sections will explain how I derived an equation for each of these 4 distributions, whereby $0 \leq d \leq 5$ and $E[d] \approx 1$, and why I decided to choose an exponential distribution in greater detail.

Linear Distribution:

A linear distribution will cause d to decrease at a constant rate. It is possible to create a linear equation for $0 \leq d \leq 5$ using the following summation:

$$\sum_{d=0}^5 (md + c) = 1$$

Since damage cannot be lower than 0 or greater than 5, the probability of any damage outside the given range will be 0. Hence, we will be able to get a linear function:

$$y = -\frac{1}{21}x + \frac{2}{7}$$

The probability distribution table will be as such:

d	$P(X = d)$
0	0.286
1	0.230
2	0.190
3	0.143
4	0.095
5	0.048

To ensure that $E[d] \approx 1$, the following formula must also be considered:

$$0a + 1b + 2c + 3d + 4e + 5f \approx 1$$

However, for given probability distribution, the $E[d] = 1.659 \rightarrow E[d] \approx 2$.
Therefore, I concluded that using linear distribution.

Quadratic Distribution:

A quadratic distribution will cause d to decrease at an increasing rate. It is possible to create a quadratic equation for $0 \leq d \leq 5$ using the following summation:

$$\sum_{d=0}^5 (ad^2 + bd + c) = 1$$

Following the method for linear distribution, I got the following quadratic equation:

$$y = -\frac{1}{125}x^2 + \frac{36}{125}$$

The probability distribution table will be as such:

d	$P(X = d)$
0	0.224
1	0.217
2	0.199
3	0.168
4	0.124
5	0.068

For this probability distribution, $E[d] = 1.955 \rightarrow E[d] \approx 2$ thus making it inappropriate for our use case.

Reciprocal Distribution:

To use reciprocal distribution, I had the following 2 conditions:

1. The probability of getting the damage 6 is 0
2. The area under the curve from $0 \leq d \leq 6$ is 1

Using these conditions, I was able to get the following equation:

$$y = -\frac{1}{x + 1.130924217} - 0.1402342767$$

However, the sum of the probabilities of d will not be 1 in this scenario. Hence, I first had to normalise the values. The equation for normalization is given below:

$$\text{Normalised Probability for a value of } d = \frac{\text{Initial value of probability for a value of } d}{\text{The sum of probability for all } d}$$

Hence, after normalisation, the probability distribution table will be as such:

d	$P(X = d)$
0	0.520
1	0.230
2	0.125
3	0.071
4	0.038
5	0.016

For this probability distribution, $E[d] = 0.925 \rightarrow E[d] \approx 1$ thus potentially making it appropriate for our use case.

Exponential Distribution:

Lastly, I also tried using exponential distribution. The formula for exponential distribution is given below²:

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

However, since we are limiting this distribution to be $0 \leq d \leq 5$, the graph greater than 5 will be 0. I also decided that the probability for which the player can get $0 \leq d \leq 3$ to be 80% hence making the critical damages i.e. 4 and 5 more rarer whilst at the same time ensuring that $E[d] \approx 1$. Using integration, I got that λ to be 0.5364793. I was also required to normalise the probabilities.

$$P(X = d) = 0.5364793e^{-(0.5364793)(x)}$$

Hence, after normalisation, the probability distribution table will be as such:

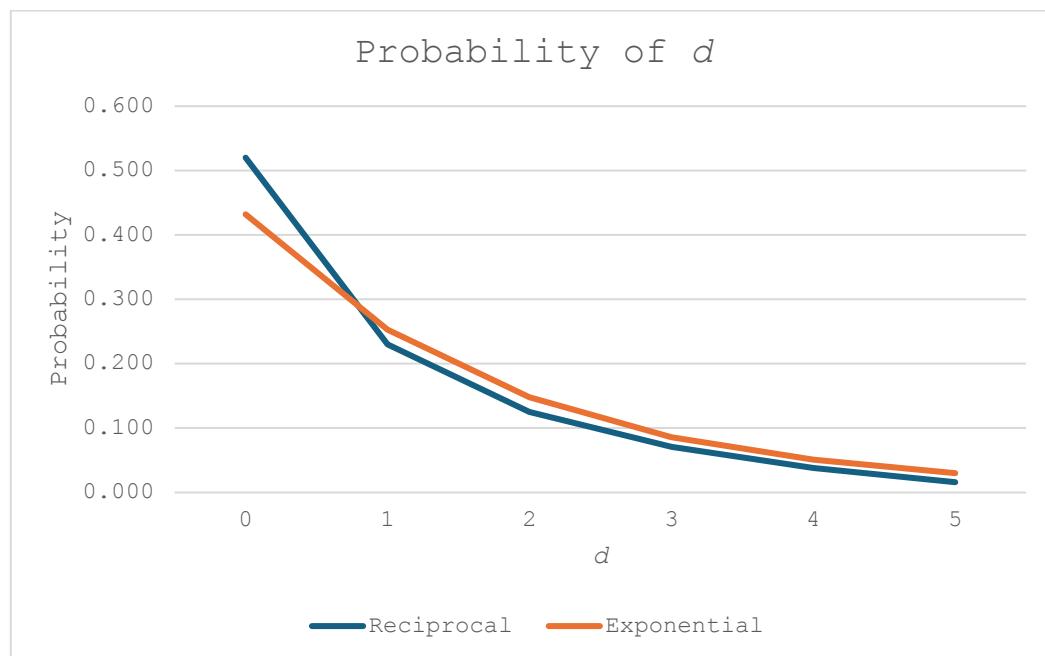
d	$P(X = d)$
0	0.432
1	0.253
2	0.148
3	0.086
4	0.051
5	0.030

² Pishro-Nik, Hossein. "Exponential Distribution | Definition | Memoryless Random Variable." *Www.Probabilitycourse.Com*, https://www.probabilitycourse.com/chapter4/4_2_2_exponential.php. Accessed 20 July 2026.

For this probability distribution, $E[d] = 1.161 \rightarrow E[d] \approx 1$ thus potentially making it appropriate for our use case.

Comparing reciprocal and quadratic distribution:

For this section d_r represents damage due to the reciprocal distribution and d_e represents the damage due to the exponential distribution. From above, we have seen that the reciprocal and quadratic distributions best match our requirements ($E[d] \approx 1$). But there are some differences between both the distributions which would affect the gameplay. Therefore, I will try to decide the better option for my game. Below is a graph of both the reciprocal distribution (blue) and exponential distribution (orange) which would help us visualise how both the distributions differ with each other:



The very first striking difference between both the distributions is that the $P(d) = 0$ for reciprocal distribution is far bigger (52%) compared to the exponential distribution (43%). This can be further highlighted by how $E[d_r] = 0.925$ whilst $E[d_e] = 1.161$. Additionally, $Std[d_r] = 1.23$ while $Std[d_e] = 1.36$. From comparing the standard deviation, of both the exponential and reciprocal distributions, the $Std[d_e] > Std[d_r]$ which means that the data is more spread out for the exponential distribution hence resulting in more variability in the data that could be chosen whilst at the same time adhering to $E[d] \approx 1$. I decided to choose the exponential distribution as a greater variability in data would result in greater uncertainty thereby making the game more unpredictable and interesting. Hence, the probability for each value of d will be following the exponential distribution table shown above.

4. Deciding the probabilities for each value of h :

For the values of h I decided to keep it simple and follow a linear model. The following criteria are meant to be followed:

- $E[h] \approx 2$
- $1 \leq h \leq 3$
- $Std[h] < 1$

The probability distribution table will look as such:

h	$P(X = h)$
1	a
2	b
3	c

The following standard deviation formula can be used to find the standard deviation of a probability distribution³:

$$\sigma = \sqrt{\sum (h - \mu)^2 \times P(X = h)}$$

Therefore, by subbing in the values, I am able to get a equation which I will label as *equation 1*:

$$1 > \sqrt{(1 - \mu)^2 \times a + (2 - \mu)^2 \times b + (3 - \mu)^2 \times c} \quad [1]$$

³ Thakur, Madhuri. "Probability Distribution Formula." *Educba*, 15 Feb. 2020, <https://www.educba.com/probability-distribution-formula/>. Accessed 5 Aug. 2026.

For the second equation, I am simply using the property that the sum of all probabilities is 1. I will label this as *equation 2*:

$$a + b + c = 1 \quad [2]$$

For the last equation, I am going to be using the mean formula as can be seen below⁴:

$$\mu = \sum [x_i \times p(x_i)]$$

The equation below is after all the values have been substituted in. I will label this as *equation 3*.

$$\mu = a + 2b + 3c \quad [3]$$

Since equation 1 and 3 both have 4 variables (including the mean), I will try to establish a value for the mean as it would make it easier to find potential values of **a**, **b** and **c**. Is it evident that mean is the same as expected value⁵ and one of the criteria is for $E[h] \approx 2$. Leaving it as an approximation will result in many different probability combinations therefore, I am going to say that $\mu = E[h] = 2$. By making this explicit, I can now further simplify equations 1 and 3.

Equation 1:

$$1 > a + c$$

Equation 3:

$$2 = a + 2b + 3c$$

Now, by making **b** the subject in *equation 2* and substituting it into *equation 3*, I get that:

$$a = c$$

⁴ Thakur, Madhuri. "Probability Distribution Formula." *Educba*, 15 Feb. 2020, <https://www.educba.com/probability-distribution-formula/>. Accessed 5 Aug. 2026.

⁵ "Mean or Expectation Value." *Gsu.Edu*, 2026, <http://hyperphysics.phy-astr.gsu.edu/hbase/Math/mean.html>. Accessed 5 Aug. 2026.

Therefore, I am going to let both $\mathbf{a} = \mathbf{c}$ be $\mathbf{x}$. Now by substituting $\mathbf{x}$ into equation 1, we can figure out that:

$$x < 0.5$$

Hence, from these conditions of $\mathbf{x}$, $\mathbf{b}$ can still range from any value from 0 to 1. Since I want the 2 to be picked the majority of the time, I decided that $\mathbf{b}$ should be higher than 0.5. However, I also wanted some form of variability and therefore I decided that $\mathbf{b} = \mathbf{0.6}$ whereby the $\mathbf{Std(h)} = \mathbf{0.6325}$. Hence, the final probability distribution that I decided for the level 0 heal function is the following:

$\mathbf{h}$	$\mathbf{P(X = h)}$
1	0.2
2	0.6
3	0.2

Part II: Creating a cooldown mechanic

I decided to introduce a cooldown mechanic in order to add more variety to the game and prevent the users from simply spamming the heal option which would result in a stalemate. I initially thought of making a cooldown mechanic whereby the user cannot use the heal option for 2 turns once they use it. However, I felt that this would simply restrict the game options and will not be as exciting. Therefore, I decided to introduce a “wounding system”.

This wounding system is created to reduce the effectiveness of the healing option thereby incentivising players to use the attack option. It also acts as a soft cooldown on the heal option thereby making players think twice before spamming the heal option. I decided that the opposing player can only be wounded when $d \geq 3$. This means that the opponent player has a 16.7% chance of being wounded. I also decided that since $E[d] \approx 1$, a wounded player's $E[h]$ must also be closer to 1. Additionally, I also decided that there should be a small chance that they will get $h = 0$. Therefore, based on these constraints, the following table shows the probability distribution for healing for every damage the player inflicts on the opponent:

$0 \geq d \geq 2$:

h	$P(X = h)$
1	0.2
2	0.6
3	0.2

$d = 3$:

h	$P(X = h)$
0	0.1
1	0.6
2	0.3

$d = 4$:

h	$P(X = h)$
0	0.3
1	0.5
2	0.2

$d = 5$:

h	$P(X = h)$
0	0.6
1	0.3
2	0.1

This “wounded” effect will last till a certain number of turns (t). I decided that $t = 3$ as it will allow for the attack to seriously impact the player’s heal mechanic.

Part III: Adding levels to the damage option

Finally, I also decided to add levels to the damage option in order to add additional incentives to use the attack option despite $E[d] < E[h]$. The level of the attack (L) option will increase passively as the user uses the attack option.

This feature will work as such:

1. Each time the player uses the attack feature, there is a probability “ $P(L)$ ”, that the player will gain one point
2. In order for the player to reach the next level of attack, the player needs to get 3 points.
3. Once the player reaches a new level of attack (i.e. Attack level 1 $\rightarrow$ Attack level 2), this point system will reset back to 0 and the player will have to repeat steps 1 and 2 in order to reach the next attack level (i.e. Attack level 3)
4. The minimum attack level will be 0 and the maximum attack level will be 4

I used the formula below to understand the expected number of turns ($E[t]$) that it will take for L to increase:

$$E[t] = \frac{3}{P(L)}$$

The table below lists the expected number of turns for each $P(L)$:

$P(L)$	$E[t]$
0.05	60.00
0.1	30.00
0.15	20.00
0.2	15.00
0.25	12.00
0.3	10.00
0.35	8.57
0.4	7.50
0.45	6.67
0.5	6.00
0.55	5.45
0.6	5.00
0.65	4.62
0.7	4.29
0.75	4.00
0.8	3.75
0.85	3.53
0.9	3.33
0.95	3.16
1	3.00

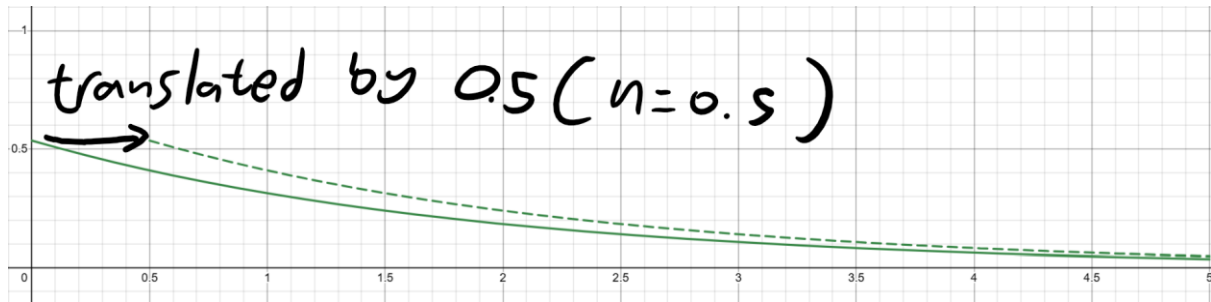
To prevent the game from exceeding beyond 100 turns, I decided that $P(L) = 0.35$. Since at level 0, $E[d] \approx 1$, both players might loose around 9hp by the time $L = 2$ if they only use the spam function (this excludes the variability of d).

For the probability distribution of each level, I decided to simply horizontally translate the exponential function that is used for d by n . The function and its range is shown below:

$$P(X = d) = 0.5364793e^{-(0.5364793)(x-n)}$$

$$n \leq x \leq 5$$

I decided that n should increase by 0.5 for each level. This would cause the graph to translate as such:



For $0 \leq x \leq n$, I used the inverse of the same function by removing the negative sign in the exponent as such:

$$P(X = d) = 0.5364793e^{(0.5364793)(x-n)}$$

Therefore, the graph for the probability distribution when $n = 0.5$ will look as such:

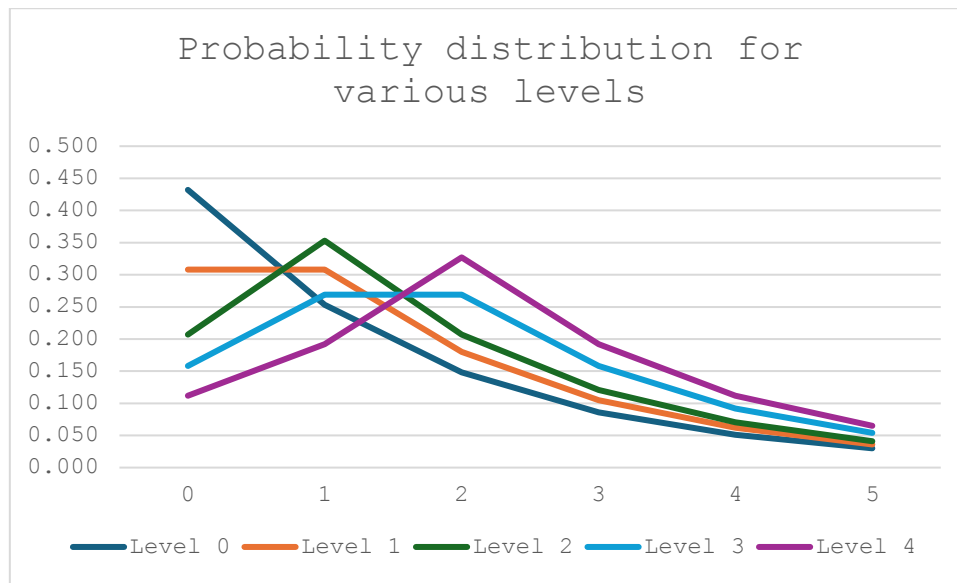


The new probabilities will still have to be normalised. The table below highlights how $E[d]$ changes with level:

Level	$E[d]$
0	1.161
1	1.411
2	1.618
3	1.919
4	2.195

Once the players reach level 4, $E[d] > E[h]$ thereby making the heal option less useful. However, with the increased standard deviation in d , and chances of wounding enemy players, it is likely that the game will end before level 4 is reached. The following table and graph show the probability distribution of the damage from level 0 to 4:

<i>d</i>	Level 0	Level 1	Level 2	Level 3	Level 4
0	0.432	0.308	0.207	0.158	0.112
1	0.253	0.308	0.353	0.269	0.192
2	0.148	0.180	0.207	0.269	0.327
3	0.086	0.105	0.121	0.158	0.192
4	0.051	0.063	0.071	0.092	0.112
5	0.030	0.036	0.041	0.054	0.065



Part IV: Is it possible to remove first-mover advantage?

It is obvious that player 1 has an advantage as seen from the payoff matrix at the start of this writeup. Although player 1 holds an initial advantage, the games stochastic nature as seen in its: ***damage done, health healed, wounding debuff*** and ***attack level ups*** helps to introduce enough variance to dampen the first mover advantage. Therefore, while it is not possible to completely remove the first-mover advantage, it has been significantly reduced.

Conclusion

The question that was initially asked was: How to create the game whereby the damage and heal output (d and h) is dependent on the skill level of the player (s) and also ensures turn-order neutrality. I managed to create a game whereby the players can and will actively attempt to use both the heal and the damage function strategically. This is done by ensuring that the standard deviation of the heal function is always lower than that of the damage function. Additionally, a level up mechanism for only the damage function and a wounding mechanism were implemented to encourage players to use the attack function so that the game will not wind up in a stalemate. By using various probability functions, the game has stochastic variance that will minimise first-player advantage thereby ensuring turn-order neutrality.

While the core model of this project addresses the question, future iterations of this project could expand on these foundations by adding additional features that will help to elevate the game.

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Citations

1. "Textual - Home." *Textual Documentation*, textual.textualize.io/.
2. Pishro-Nik, Hossein. "Exponential Distribution | Definition | Memoryless Random Variable." *Www.Probabilitycourse.Com*,
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